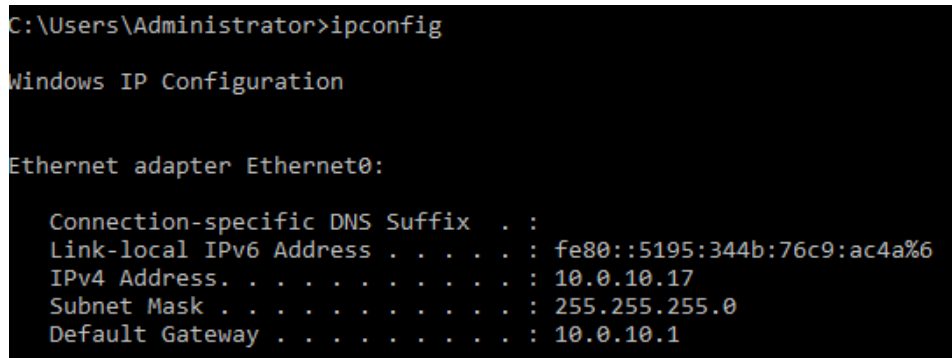


Operating system configurations

The Windows Server 2016 ISO was downloaded from the Microsoft Evaluation Center. The network adapter of the virtual machine for the Windows Server 2016 Datacenter was assigned the vmnet3 before the Windows Active Directory Domain Controller was deployed. 5 GB of RAM and 60 GB of disk space were reserved for the guest. The IP address of the domain controller was defined as 10.0.10.17, the default gateway as 10.0.10.1 and its DNS server as 127.0.0.1.

The IP configuration was verified using the output of the command `ipconfig`.



```
C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::5195:344b:76c9:ac4a%6
    IPv4 Address. . . . . : 10.0.10.17
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.10.1
```

Figure 1: Verification of IP configuration on the Windows Server 2016 guest

The Windows 10 Education, version 22H2 ISO was downloaded from the Microsoft Azure Developer Tools student account. The virtual machines for the two Windows 10 clients were configured with the vmnet4. 2 GB of memory and 60 GB of disk space were designated for each guest. The IP addresses on the two Windows 10 machines were set as 10.0.17.3 and 10.0.17.4, their default gateways as 10.0.17.1 and the DNS server as 10.0.10.17.

,

The output of the command `ipconfig` was used to validate the IP configuration on both Windows clients.

```
C:\Users\yulia>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::9443:29bf:e5bb:7f4f%2
    IPv4 Address. . . . . : 10.0.17.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.17.1
```

Figure 2: Verification of IP configuration on the first Windows 10 guest

The two Windows 10 machines were configured to join the domain, and it was verified that each machine was a member of the domain by browsing to Active Directory Users and Computers > ec.sa.com > Computers section, where the host names of the Windows 10 domain members were listed.

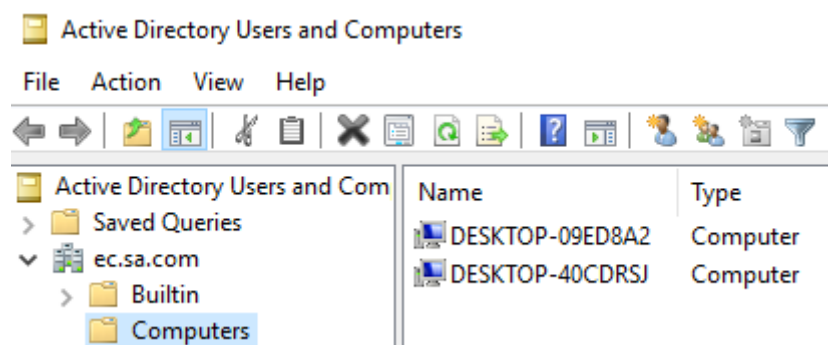


Figure 3: Verification of the two Windows 10 machines joining the domain in the Active Directory Users and Computers section

In a deviation from the plan, the two Windows 10 machines running version 1903 were downloaded from the Internet Archive and reinstalled in the simulated environment. This release was necessary for the two persistence exploits in the attack chain to work.